

Build a Web App: GitHub Codebase Personality Profiler

Project Overview

Build a full-stack web application called **Codebase Personality Profiler**.

The application analyzes a GitHub repository's commit history, contributor behavior, commit messages, file churn patterns, and development habits to generate a humorous but evidence-based "personality profile" of the codebase and its team.

The goal is not to produce a generic repository summary. The goal is to generate a narrative that feels like a psychological profile, horoscope, or character analysis while remaining grounded in measurable repository data.

Every claim made by the AI must be traceable to supporting metrics and source commits.

Think:

"This repository is anxious. 42% of commits contain the words 'fix', 'bug', or 'hotfix'. Most of those commits happen after 9 PM, suggesting a tendency toward last-minute firefighting."

instead of:

"This repository focuses on bug fixing."

The tone should be clever, witty, memorable, and slightly uncomfortably accurate.

Core User Flow

1. User visits homepage.
2. User enters:
 3. GitHub repository URL
 4. or owner/repository name
5. User clicks Analyze.
6. Backend fetches repository data using GitHub API.
7. Backend computes repository metrics.
8. Backend sends structured metrics and commit examples to an LLM.

9. LLM generates personality profile.
 10. Frontend displays:
 11. Personality report
 12. Supporting evidence
 13. Visualizations
 14. Contributor insights
 15. Download/share options
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Tech Stack

Frontend:

- Next.js
- React
- TypeScript
- Tailwind CSS
- Recharts

Backend:

- Node.js
- Express or Next.js API routes
- TypeScript

Database:

- PostgreSQL

ORM:

- Prisma

AI:

- OpenAI API

Deployment:

- Vercel

Authentication:

- GitHub OAuth (optional)

Data Collection Layer

Use GitHub REST API and GraphQL API.

Collect:

Repository Metadata

- Repository age
- Stars
- Forks
- Open issues
- Closed issues
- Contributors

Commit History

For every commit:

- SHA
- Author
- Timestamp
- Message
- Files changed
- Additions
- Deletions
- Total diff size

Pull Requests

Collect:

- Opened date
 - Merged date
 - Review count
 - Time to merge
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Contributors

Collect:

- Number of commits
 - Active days
 - Active hours
 - File ownership patterns
-

Metrics Engine

Compute the following metrics.

Activity Metrics

Commit Frequency

- commits per day
- commits per week
- commits per month

Time of Day

Distribution:

- morning
- afternoon
- evening
- night

Determine:

- night owl score
 - healthy work schedule score
-

Stability Metrics

Calculate:

Fix Ratio

Percentage of commits containing:

- fix
- bug
- hotfix
- patch

Feature Ratio

Percentage containing:

- feat
- feature
- add

Refactor Ratio

Percentage containing:

- refactor
- cleanup
- restructure

Test Ratio

Commits touching:

- tests
- specs
- testing folders

Churn Metrics

Measure:

File Volatility

Most frequently modified files.

Churn Score

How often files are rewritten.

Rewrite Frequency

Number of times files experience major edits.

Deadline Behavior

Detect:

- large commits before releases
- bursts of activity
- commit spikes

Generate:

- procrastination score
 - deadline panic score
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Team Dynamics

Measure:

Bus Factor

Number of people responsible for majority of code.

Ownership Concentration

Whether one person dominates.

Collaboration Index

Number of shared files edited by multiple people.

Lone Wolf Score

How siloed contributors are.

Testing Discipline

Measure:

- code changes with tests

- code changes without tests
- ratio

Generate:

- optimism score
 - caution score
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Personality Trait System

Create traits derived from metrics.

Examples:

Anxious

Triggered by:

- high bug-fix ratio
 - high after-hours activity
 - high churn
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Perfectionist

Triggered by:

- many refactors
 - frequent cleanup commits
 - extensive testing
-

Chaotic

Triggered by:

- large commit spikes
 - inconsistent schedules
 - huge diffs
-

Disciplined

Triggered by:

- stable cadence
 - consistent testing
 - small commits
-

Optimistic

Triggered by:

- lots of feature commits
 - low testing ratio
-

Paranoid

Triggered by:

- extensive testing
 - many validation changes
-

Workaholic

Triggered by:

- night commits
 - weekend commits
-

Architect

Triggered by:

- frequent refactoring
 - strong structure evolution
-

LLM Narrative Generation

The most important part.

The AI should NOT invent traits.

The backend must provide:

- metric values
- trait scores
- supporting commit examples
- notable files

Prompt the LLM:

"You are a witty software psychologist.

Analyze repository behavior.

Write like a mix of:

- a therapist
- a workplace consultant
- a horoscope writer
- a stand-up comedian

Never invent evidence.

Every personality claim must be supported by supplied metrics.

For each personality trait:

1. State trait.
2. Explain behavior.
3. Reference evidence.
4. Include one memorable metaphor.

Be funny but accurate."

Explainability Requirement

Every personality claim must contain:

Trait

Example:

Anxious

Why

42% of commits are fixes.

Evidence

- Fix login race condition
- Hotfix production issue
- Patch auth bug

Metrics

- Fix ratio: 42%
- Night commits: 31%

This should be displayed in expandable cards.

Visualizations

Create charts:

Commit Activity Heatmap

GitHub-style heatmap.

Commit Time Distribution

Bar chart.

Commit Type Breakdown

Pie chart.

Contributor Activity

Stacked chart.

File Churn Ranking

Horizontal bar chart.

Trait Radar Chart

Axes:

- Anxiety
 - Discipline
 - Chaos
 - Optimism
 - Perfectionism
 - Collaboration
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Personality Report Layout

Header

Repository Name

Overall Personality

Example:

"The Anxious Perfectionist"

Personality Summary

3–5 paragraphs.

Dominant Traits

Cards with:

- trait
 - score
 - evidence
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Team Dynamics

Narrative analysis.

Development Habits

Narrative analysis.

Most Dramatic Commit

Identify commit with:

largest diff or most emotional message

Display separately.

Fun Statistics

Examples:

- Most active hour
 - Most active contributor
 - Most edited file
 - Longest commit message
 - Largest commit
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Sharing Features

Generate:

Personality Card

Shareable image.

Contains:

- repository name
 - personality type
 - trait radar
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Public Link

Permanent report URL.

Stretch Features

1. Compare two repositories.
 2. Compare personality over time.
 3. Contributor personality profiles.
 4. Team compatibility analysis.
 5. Repository evolution timeline.
 6. Roast mode.
 7. Executive summary mode.
 8. Recruiter mode.
 9. Founder mode.
 10. AI-generated repository horoscope.
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Design Direction

Style should feel like:

- GitHub wrapped
- Spotify Wrapped
- Personality tests
- Data journalism

Avoid corporate dashboards.

Prioritize delight, humor, and storytelling.

Users should want to share their results on social media.